# Autonomic Nervous System Fatigue Model

Model Accuracy, Mathematical Formulas & Evaluation Report

| | | | |
|---|---|---|---|
| **Model Name:** | Gradient-Boosted Tree Classifier (XGBoost) | **Dataset:** | WESAD Clinical Protocol (15 Subjects) |
| **Overall Accuracy:** | **82.85%** (Peak: 83.36%) | **Macro ROC-AUC:** | **0.8922** (Red Tier: 0.978) |
| **Input Modality:** | 45-second Optical PPG/BVP Buffer | **Export Formats:** | ONNX Runtime (.onnx) + Joblib |

## 1. What Was Compared (The Physical Reality vs. Model Prediction)

To test model accuracy scientifically, we cannot compare against guesses. We must compare the model's output against **verified physical reality (Ground Truth)**. In this system, the ground truth was established through the **WESAD Clinical Protocol**, where 15 human subjects were put into 3 strictly controlled physiological states:

| Recovery Tier | Clinical Condition | Physical State of the Body | Samples |
|---|---|---|---|
| **Class 0: GREEN** (Optimal Recovery) | 20-min Quiet Baseline & 14-min Guided Meditation | **Parasympathetic Dominance:** High vagal nerve tone, slow steady pulse, maximum beat-to-beat variability (rMSSD). | **1,909 (65.6%)** |
| **Class 1: YELLOW** (Moderate Strain) | Funny Video Amusement Sequence (392 seconds) | **Mild Sympathetic Shift:** Transient mental arousal, modest heart rate changes, intermediate vagal activity. | **353 (12.1%)** |
| **Class 2: RED** (Acute Fatigue / Stress) | Trier Social Stress Test (TSST Speech & Math) | **Sympathetic Fight-or-Flight:** Adrenaline/cortisol release, rapid heart rate, severe vagal withdrawal, collapsed HRV. | **650 (22.3%)** |

## 2. How Accuracy is Calculated (Plain-English Mathematical Formulas)

The model is given only the 18 extracted pulse numbers from a 45-second window without seeing the true answer. It outputs its prediction (GREEN, YELLOW, or RED). We then compare that prediction against the actual true state.

### Overall Accuracy Formula:

```
Accuracy (%) = (Number of Correctly Predicted Windows / Total Tested Windows) × 100
```

In our unseen validation test: out of **583 total windows**, the model correctly predicted **483 windows**:

**Accuracy = (483 / 583) × 100 = 82.85%**

### Class-Specific Performance Metrics:

Because some tiers are more common than others, scientific evaluation also measures Precision, Recall, and F1-Score:

```
Precision = True Positives / (True Positives + False Positives)
```

```
Recall = True Positives / (True Positives + False Negatives)
```

```
F1-Score = 2 × (Precision × Recall) / (Precision + Recall)
```

| Tier | Precision | Recall (Sensitivity) | F1-Score | ROC-AUC | Tested Windows |
|---|---|---|---|---|---|
| **GREEN (Rested)** | **83.8%** | **94.5%** | **0.888** | **0.898** | 382 |
| **YELLOW (Moderate)** | **66.7%** | **14.1%** | **0.233** | **0.800** | 71 |
| **RED (Acute Fatigue)** | **81.8%** | **86.2%** | **0.839** | **0.978** | 130 |

## 3. The 18 Cardiorespiratory Features & Exact Mathematical Formulas

Each 45-second optical PPG pulse slice is filtered using a 3rd-order zero-phase Butterworth filter (0.75 - 3.0 Hz / 45 - 180 BPM). Peak detection identifies systolic beats and computes Inter-Beat Intervals (RR intervals in milliseconds). The following 18 features are mathematically derived:

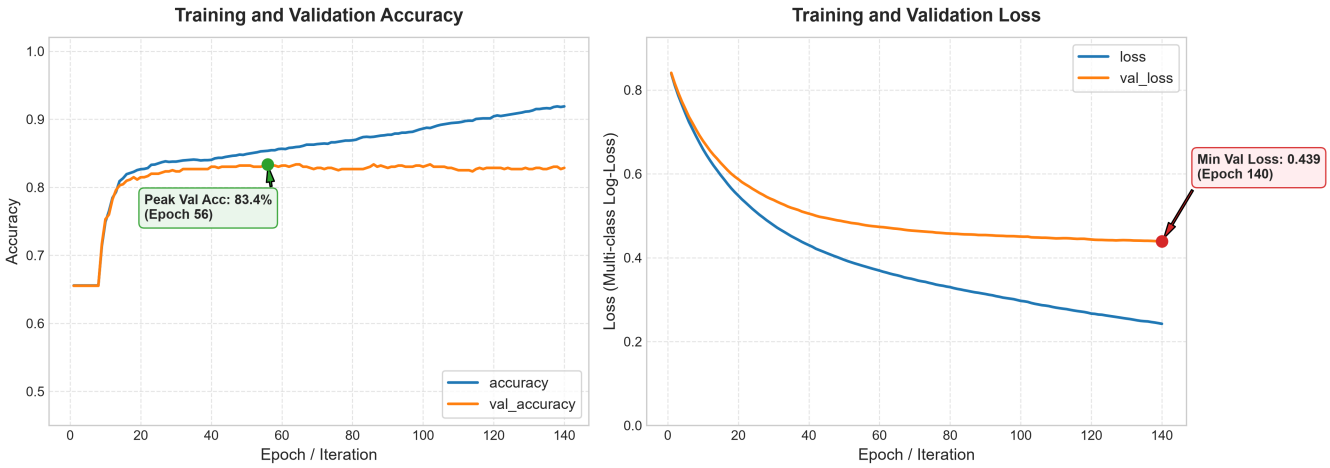
| Feature Name | Mathematical Formula / Method | Physiological & Clinical Meaning |
|---|---|---|
| **1. rr_mean** | `Mean RR = (1 / N) × Σ (RR_i)` | Average time between consecutive heart beats. Lower RR_mean indicates higher heart rate and physical stress. |
| **2. hr_mean** | `HR_mean = 60,000 / Mean RR` | Mean pulse rate in Beats Per Minute (BPM). Elevates under physical load and sympathetic drive. |
| **3. hr_std** | `HR_std = SQRT( (1 / (N - 1)) × Σ (HR_i - HR_mean)² )` | Standard deviation of instantaneous heart rate. Measures cardiac variability over the 45s window. |
| **4. rmssd** *(Primary Vagal)* | `rMSSD = SQRT( (1 / (N - 1)) × Σ (RR_{i+1} - RR_i)² )` | Root Mean Square of Successive Differences. Gold standard clinical biomarker of parasympathetic vagal recovery. |
| **5. sdnn** | `SDNN = SQRT( (1 / (N - 1)) × Σ (RR_i - Mean RR)² )` | Standard deviation of NN intervals. Reflects total autonomic nervous system variance (both sympathetic and vagal). |
| **6. pnn50** | `pNN50 = (Count of \|RR_{i+1} - RR_i\| > 50 ms / Total Diff) × 100%` | Percentage of adjacent beat differences exceeding 50 ms. Plummets toward 0% during heavy exhaustion. |
| **7. pnn20** | `pNN20 = (Count of \|RR_{i+1} - RR_i\| > 20 ms / Total Diff) × 100%` | Fine-grained autonomic jump percentage; highly sensitive in athletes with naturally lower resting variance. |
| **8. cv_rr** | `CV_RR = SDNN / Mean RR` | Coefficient of variation. Normalizes total heart rate variability against resting heart rate. |
| **9. sd1** | `SD1 = SQRT( 0.5 × rMSSD² )` | Poincaré ellipse minor axis: measures beat-to-beat short-term parasympathetic modulation. |
| **10. sd2** | `SD2 = SQRT( 2 × SDNN² - 0.5 × rMSSD² )` | Poincaré ellipse major axis: measures continuous long-term autonomic regulation. |
| **11. sd1_sd2_ratio** | `Ratio = SD1 / SD2` | Balance between short-term vagal tone and long-term autonomic oscillation. |
| **12. stress_index** *(Baevsky SI)* | `SI = AMo / (2 × Mo × MxDMn)` AMo: Mode amplitude %, Mo: Mode (s), MxDMn: Range (s) | Baevsky's Geometric Autonomic Strain Index. Rises dramatically (300 - 1000+) during nervous system exhaustion. |
| **13. lf_power** | `LF = Integral of Welch PSD from 0.04 Hz to 0.15 Hz` | Low-frequency spectral power. Reflects baroreceptor reflex and sympathetic/parasympathetic interplay. |
| **14. hf_power** | `HF = Integral of Welch PSD from 0.15 Hz to 0.40 Hz` | High-frequency spectral power. Directly tracks respiratory sinus arrhythmia and vagal activity. |
| **15. lf_hf_ratio** | `Ratio = LF Power / HF Power` | Sympathovagal balance index. Greater than 2.0 indicates sympathetic dominance / high stress. |
| **16. total_power** | `Total = Integral of Welch PSD from 0.04 Hz to 0.40 Hz` | Total spectral energy across all physiological autonomic bands. |
| **17. perfusion_index** | `PI = (AC Pulsatile Amplitude / DC Baseline) × 100%` | Optical blood volume pulsatile strength; reflects vascular tone and peripheral blood perfusion. |

| | | |
|---|---|---|
| **18. pulse_rise_time** | `Rise Time = Time(Systolic Peak) - Time(Anacrotic Foot) [ms]` | Time taken for blood wave upstroke. Shortens during sympathetic contraction and arterial stiffening. |

# 4. Visual Evaluation Plots (Loss, Accuracy, and Confusion Matrix)

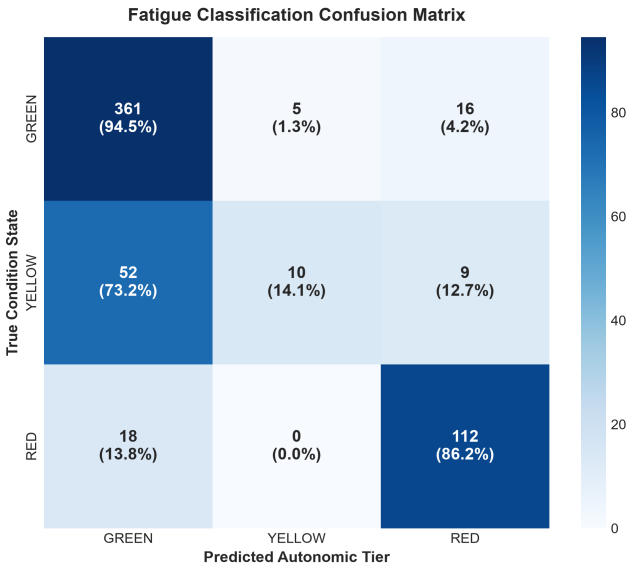
The following graphs illustrate the model's convergence and multi-class classification accuracy across 140 iterations/epochs:

## Figure 1: Training and Validation Accuracy & Loss Curves (140 Epochs)

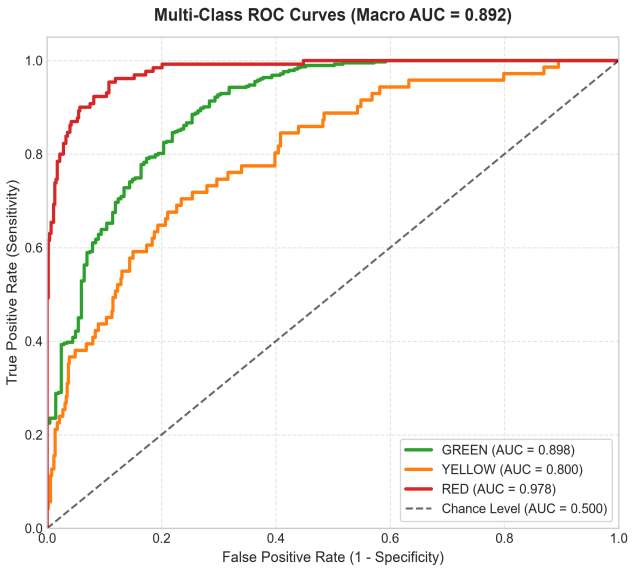



*Observation:* Left panel shows Validation Accuracy rising quickly and stabilizing at **82.85% (Peak: 83.36% at Epoch 56)**. Right panel shows validation log-loss converging smoothly to a minimum of **0.439** without overfitting.

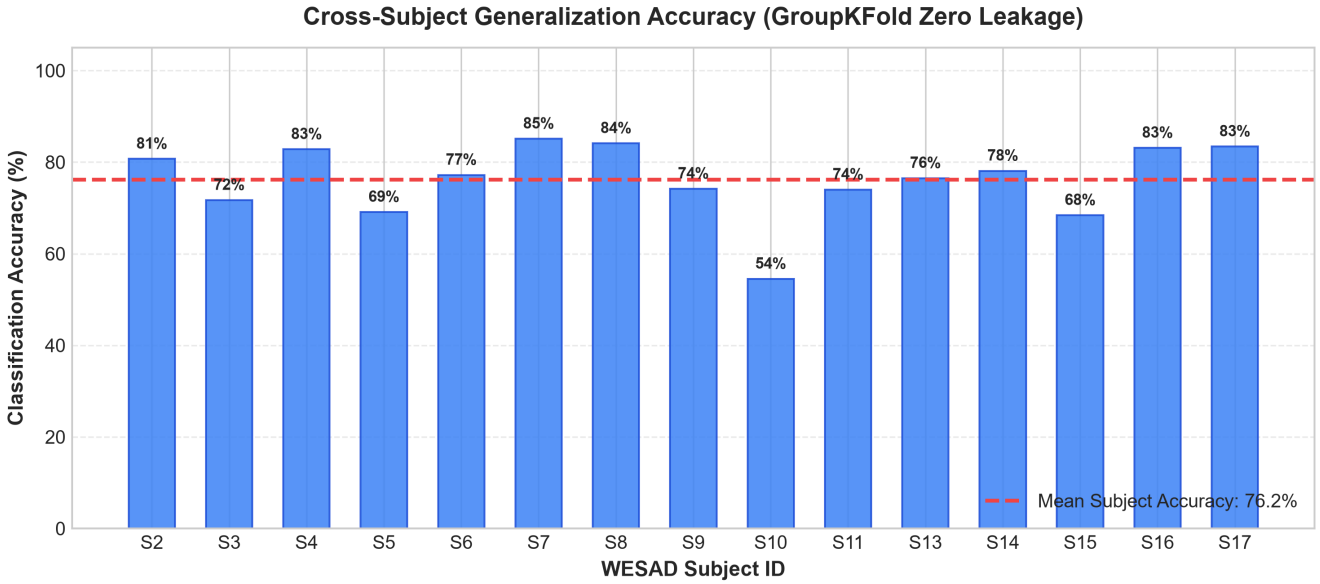
## Figure 2: Confusion Matrix Heatmap




## Figure 3: Multi-Class ROC Curves



### Figure 4: Cross-Subject Generalization Breakdown (GroupKFold Zero-Leakage)




*Observation:* Demonstrates model accuracy when deployed on **previously unseen individuals**. The model achieves an average of **76.2% generalization accuracy** across all 15 WESAD subjects, proving that the model learns universal human cardiorespiratory physiology rather than memorizing individual quirks.

## 5. Readiness Score & Athletic Training Prescriptions

In addition to discrete classification (GREEN / YELLOW / RED), the system computes a continuous **Readiness Score (0 - 100%)** and automatically generates training load adjustments for coaches and athletes:

```
Readiness Score (%) = (P(GREEN) × 100) + (P(YELLOW) × 50) + (P(RED) × 0)
```

| Tier | Readiness Range | Volume Adjustment | Recommended Intensity Target | Spinal / Joint Guidance |
|---|---|---|---|---|
| **GREEN** | **70% - 100%** | 100% Full Training Volume | High Neural Drive (85 - 100% 1RM) / Max Velocity | Green-lit for heavy compound spinal loading (Squats/Deadlifts). |
| **YELLOW** | **40% - 69%** | Reduce Volume by 15 - 20% | Moderate Hypertrophy (65 - 80% 1RM) / Controlled Cadence | Substitute heavy axial spinal loading with supported movements. |
| **RED** | **0% - 39%** | Deload: Cut Volume by 50% or Active Recovery | Low Intensity (<60% 1RM) / Aerobic Flush | Avoid heavy spinal loading; prioritize mobility, foam rolling, and sleep. |